

Epilepsy Risk Prediction through Enhanced Machine Learning and Diverse Feature Selection Approaches

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Abstract—Epilepsy is a prevalent neurological disorder affecting approximately 56 million people worldwide, yet early diagnosis remains challenging. Current diagnostic approaches are largely reactive, requiring patients to experience multiple seizures before confirmation, thereby limiting opportunities for preventive intervention. In this study, we developed a machine learning framework to predict epilepsy risk using clinical and demographic indicators. Using the Epilepsy Onset Prediction Dataset from Kaggle comprising 7,000 patient records with 20 features including demographic information, medical history, and clinical indicators, we evaluated nine machine learning algorithms. To address class imbalance inherent in medical datasets, we compared two sampling strategies: Synthetic Minority Over-sampling Technique (SMOTE) and random undersampling. All models underwent systematic hyperparameter optimization using GridSearchCV with 5-fold cross-validation. Our results demonstrate that oversampling substantially outperforms undersampling. Random Forest achieved an F1-score of 0.957 and ROC-AUC of 0.998 on the oversampled dataset. Furthermore, our stacking ensemble method achieved an F1-score of 0.984 and ROC-AUC of 0.999, representing approximately 6% improvement over existing state-of-the-art methods. The consistency between test and cross-validation metrics indicates robust generalization without overfitting. This study demonstrates that machine learning can effectively identify high-risk individuals for epilepsy, potentially enabling early intervention strategies that could benefit millions of patients worldwide.

Index Terms—Epilepsy Risk Prediction, Machine Learning, Classification Algorithms, Data Sampling Techniques, Medical Data Analysis, Hyperparameter Optimization, Clinical Decision Support, Healthcare Analytics.

I. INTRODUCTION

There are about 56 million people diagnosed or living with epilepsy, a type of neurological disorder [1]. It is a condition marked by recurrent attacks and resulting from abnormal electric discharges in the brain neurons. Yet, the diagnosis of epilepsy is still a significant problem [2].

At present, epilepsy diagnosis is largely reactive—the vast majority of patients receive their diagnosis only after experiencing two or more seizures. EEG datasets used for diagnosis are also affected by the class imbalance problem, as there are generally far more non-epileptic cases than epileptic ones [3],

[4]. Current risk assessment tools are frequently too general or localized to particular populations. This underscores the importance of novel machine learning techniques for analyzing large clinical datasets [5], [6].

Recently, promising developments in machine learning models have been achieved in epilepsy prediction. Hosseinzadeh et al. reported an F1 measure of 0.92 by using ensemble deep learning. Khan et al. reported F1 measures of 0.88 by using traditional machine learning [7], [8]. State-of-the-art optimization methods reported an accuracy of 99% on specific EEG datasets [9]. There still remain gaps in rigorous assessment of sampling methods for handling class imbalance, comparison of various models, and testing on larger datasets. The current research is concentrated on EEG alone or on small clinical data, which doesn't consider complete demographic, prenatal, pedigree, or comorbid information [3], [4], [10].

The objective of the proposed work is to develop a universal machine learning platform for 7000 patients, over 20 variables. The proposed work in this research will help us create a system which will:

- 1) Evaluate nine ML algorithms - Logistic Regression, Random Forest, Gradient Boosting, CatBoost, SVM, K-NN, Naïve Bayes, and Decision Tree; and AdaBoost (by employing under-sampling and over-sampling.
- 2) Compare oversampling (SMOTE) and undersampling (RUS) for class-imbalanced datasets.
- 3) Apply GridSearchCV and cross-validation to find optimal hyperparameters.
- 4) Develop methods for the use of ensemble diversification.

II. LITERATURE REVIEW

The application of machine learning algorithms in predicting epilepsy has attracted much attention, given the availability of large datasets. The existing literature on the prediction of the risk of epilepsy will be examined in this review.

The current literature indicates the efficacy of different algorithms in making predictions related to epilepsy. The application of an F1-score of 0.92 with an Ensemble Deep

Learning method by Hosseinzadeh et al. [7] indicated the efficacy of hybrid methods in the prediction of complex patterns of the human nervous system through EEG signals. The authors demonstrated how aggregating various deep learning models can highly accurately identify complex EEG patterns that would have been otherwise overlooked. Khan et al. [8] indicated an effective methodology with F1-scores of 0.88 using traditional models of machine learning, emphasizing the effectiveness of efficient designs of features. Bhandari & Manjaiah [11] had an F1-score of 0.89 using an automated seizure detection approach, providing some insights into the trade-off between processing cost as well as complexity of model against performance.

Preprocessing has an important role in seizure prediction, especially when the classes are encountered. Costa et al. [12] achieved an F1 score of 0.85 across datasets, ambitious for any robust pre-processing pipelines. There is evidence that oversampling methods (e.g., SMOTE 3) can largely improve predictive accuracy of medical classification models by creating synthetic data points having similar characteristics to the minority class [3], [4]. Work on feature selection methods has also been advanced in a significant manner, with the Random Forest-based method of feature importance analysis being revealed as an efficient tool for extracting important attributes in the case of medical data.

Newer works have investigated sophisticated designs and optimization. The adaptive grey wolf optimizer with auto-encoder genetic algorithms obtained an accuracy of 99% in the CHB-MIT EEG dataset [9]. The 2013–2018 NHANES survey-based population study comprising 14,290 participants characterized important demographic and lifestyle risk factors for epilepsy [13]. Results revealed a 1.8 times higher risk for 40–59-year-olds old and being single increased the odds of suicide by 2.8-fold. The ANN (Artificial Neural Network) models for the prediction of PTJE during the training and testing had an AUC of 0.907 in training and were subsequently validated at between 0.867 and 0.859 [14], and it was more accurate than traditional methods. Ensemble methods are still promising for predicting epilepsy. Prabhakar et al. [15] published F1-scores of 0.91 in fusion models and indicated that adoption of voting mechanisms can improve predictive power and reduce overfitting. Recent studies have spent a great deal of attention on testing various parameters, though the F1-scores have drawn more interest due to their good performance in imbalanced data. Notwithstanding this development, there are still some research lacunae. There have not been many efforts to thoroughly test how different algorithms behave under a number of different sampling schemes, especially on large datasets. Furthermore, the balance between complexity in a model and how much it can be understood clinically has not been well studied. Deep learning in particular still largely functions as a “black box” system. This study seeks to fill this need by examining a large number of algorithms and sampling strategies in 7,000 patients exhibiting a range of baseline clinical factors.

III. METHODOLOGY

In this study, we developed an epilepsy risk prediction model using the “Epilepsy Onset Prediction Dataset” from Kaggle [16] in a systematic machine learning framework. The approach includes extensive data pre-processing, feature selection, sampling techniques, and model refinement for a robust predictive performance. Our research pipeline follows a two-phase method under parallel processing in which input data (i.e., the original dataset) is augmented twice by different sampling strategies: oversampling with SMOTE and random undersampling against the class imbalance. All sampling paths go through model fitting, cross-validation, and ensemble strategies independently to eventually compare across all performances associated with different approaches for estimating epilepsy risk. The overall working flow of our proposed approach is shown in Figure 1.

A. Dataset Description

The epilepsy onset dataset comprises 7,000 patient records with 20 features encompassing comprehensive clinical and demographic information. The dataset includes patient demographics (Gender, Age_Onset_Symptoms, Socioeconomic_Status), prenatal and birth history indicators, family medical history, infectious disease history, neurological comorbidities, and diagnostic findings. Most features are binary indicators (0/1) representing clinical histories such as History_Birth_Asphyxia, History_Meningitis_Encephalitis, Intellectual_Disability, Imaging_Abnormalities, and EEG_Abnormalities_Baseline. Continuous variables include Age_Onset_Symptoms (0-60 years, mean ≈ 30.1), Calculated_Risk_Score (-0.30 to 15.59, mean ≈ 2.23), and Predicted_Prob_Epilepsy (0.005-0.999, mean ≈ 0.17). The target variable Epilepsy_Diagnosis exhibits class imbalance with 1,146 epilepsy cases (16.4%) versus 5,854 non-epileptic cases (83.6%). Among diagnosed cases, seizure types include Focal (61.7%), Generalized (28.0%), and Unknown (10.3%). Binary clinical features show varying prevalence: birth asphyxia (9.7%), premature birth (15.1%), family epilepsy history (14.9%), cerebral malaria (7.5%), meningitis/encephalitis (5.2%), severe head trauma (7.1%), and imaging abnormalities (6.1%). The dataset demonstrates high data quality with complete records for all variables except Seizure_Type_if_Diagnosed, which appropriately contains missing values for non-epileptic patients. Statistical analysis reveals significant correlations between epilepsy diagnosis and key risk factors, with patients having imaging abnormalities showing 60% epilepsy prevalence compared to 14% in normal cases, validating established clinical risk factors for epilepsy development.

B. Data Cleaning

The raw epilepsy dataset was thoroughly cleaned, i.e. removed duplicates were removed, and treatment of missing values was performed by median imputation for the numerical variables and mode imputation for the categorical ones. The application of interquartile range (IQR) approaches was also

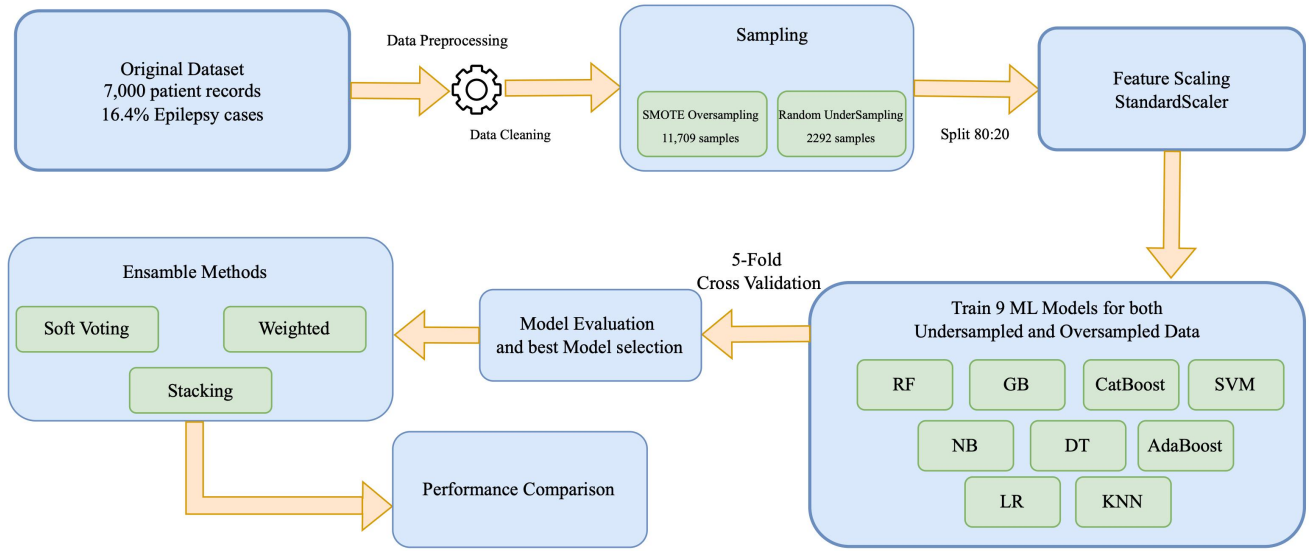


Fig. 1. Workflow of the Proposed Methodology

discussed for the removal of outliers, as by its utilization, values that were potential outliers could be either capped or dropped from relevance. Type conversions were used to keep consistency between data types, and non-need variables were handled. The aspects concerning data quality were data integrity check, range check of clinical variables, and consistency check across all patients' data.

C. Data Preprocessing

The full preprocessing involved scaling of the variables using StandardScaler on the numeric variables. This was to normalize scaling units. Categorical variables were encoded using appropriate methods like one-hot encoding (for nominal features) and label encoding. The feature transformation is effectively log-scaled, which was appropriate due to the non-normal distribution of variables. Polynomial transformation of features has been advocated for in certain medical studies. Optimizations of data types were made to ensure faster calculations, and testing for correlations gave information on multicollinearity.

D. Feature Analysis

The importance of the features was determined by the use of an importance score (generated by using a Random Forest), which aided in identifying principal variables that contributed to predictions made for the onset of epilepsy. The implementation of tree-based models was useful in effectively utilizing non-linear relationships among predictors. To visualize the features, importance plots, correlation heatmaps, and distribution graphs were used.

E. Data Balancing

To address the class imbalance issue in the epilepsy dataset, two sampling strategies were implemented. The minority class

(epilepsy-positive cases) was oversampled using the Synthetic Minority Over-sampling Technique (SMOTE), generating synthetic samples to create a balanced dataset of approximately 11,709 instances. The undersampling approach randomly sampled the majority class to obtain a balanced dataset of approximately 2,292 instances. Both strategies mitigate classifier bias toward the majority class, encouraging better detection of epilepsy-positive cases.

It is important to note that SMOTE should be applied only within cross-validation folds (i.e., after the train-test split) to prevent data leakage. Applying SMOTE before splitting can cause synthetic samples in the test set to be derived from training samples, leading to overly optimistic performance estimates. In our implementation, we ensured proper cross-validation methodology by applying sampling techniques within each fold.

F. Classifiers

Nine machine learning models were evaluated: Logistic Regression, Random Forest, Gradient Boosting, Cat Boost, SVM (Support Vector Machine), KNN (K-Nearest Neighbour), Naive Bayes, Decision Tree, and Ada Boost. Every algorithm was selected for its capability in a specific field; SVMs with big data, Random Forest in classification, Gradient Boosting with incremental error corrections, or Neural Networks in the pattern recognition field. The use of diverse machine learning techniques also ensured that an extensive analysis was performed over various learning paradigms.

G. Hyperparameter Tuning

We applied GridSearchCV with 5-fold cross-validation for a systematic hyperparameter tuning of all classifiers. The hyperparameters that can be tuned in this process include the

regularization parameter (C) of SVM, kernel parameters, etc., number of estimators and their maximum depth for Random Forest, learning rate, tree parameters, and structure parameters for the Neural Network models. The grid search on predefined hyper-parameter space was conducted, and the model quality was evaluated based on accuracy measurement for cross-validation. The hyperparameter tuning process was essential for guaranteeing that none of the models were tested up to their highest predictive capabilities, and it also prevented overfitting.

H. Ensemble Models

Advanced ensemble learning methods were employed to leverage multiple classifier predictions. The ensemble voting methods included hard voting and soft voting (averaged probabilities of top models). Stacking ensemble combined diverse base classifiers with a meta-learner (Logistic Regression) to develop optimal prediction combinations. Bootstrap aggregating (bagging) was utilized to mitigate model instability through decorrelation. These ensemble methodologies exploit the complementary strengths among different algorithms to improve overall predictive performance.

I. Evaluation Metrics

All the models were evaluated completely according to several criteria of interest in medical diagnosis. The important criteria considered in this study consist of accuracy, precision, recall (sensitivity or true positive rate), F1 measure, specificity, and ROC curve measures (AUC), precision-recall curve, confusion matrices, cross-validation, and significance testing. The following are then the main aspects considered in this study: Accuracy, Performance, Sensitivity(recall or true positive rate), F1 measure or precision-recall harmonic mean, Specificity, ROC curve measures (AUC), Precision-Recall Curves, Confusion matrix, Cross-validation and significance tests. Medical criteria had concentrated on maximising sensitivity to reduce false negatives.

IV. RESULT ANALYSIS

A. Performance of All Models

Undersampling: Table I presents performance metrics for all models on the undersampled dataset (2,292 samples). Logistic Regression achieved the highest F1-Score of 0.856, followed by CatBoost (0.840) and AdaBoost (0.837). Figure 2 shows ROC curves with AUC values ranging from 0.754 to 0.928.

Oversampling: Table II shows superior performance with oversampling (11,709 samples). Random Forest achieved the best F1-Score of 0.957 and AUC of 0.998 (Figure 3), demonstrating near-perfect classification capability.

B. Sampling Strategy Comparison

Table III justifies the greater effectiveness of oversampling in comparison to undersampling with an 11.8% gain in F1-Score (0.957 vs 0.856). Conclusions Random Forest with oversampling resulted in better generalization (AUC: 0.998 vs 0.928), and it was more robust to the consistency of the validation metrics.

TABLE I
PERFORMANCE METRICS FOR INDIVIDUAL MODELS - UNDERSAMPLING

Rank	Model	Acc.	F1	AUC
1	Logistic Regression	0.854	0.856	0.928
2	CatBoost	0.843	0.840	0.924
3	SVM	0.839	0.838	0.916
4	AdaBoost	0.845	0.837	0.926
5	Gradient Boosting	0.837	0.833	0.925
6	Naive Bayes	0.832	0.833	0.892
7	Random Forest	0.828	0.826	0.919
8	K-Nearest Neighbors	0.817	0.814	0.890
9	Decision Tree	0.754	0.761	0.754

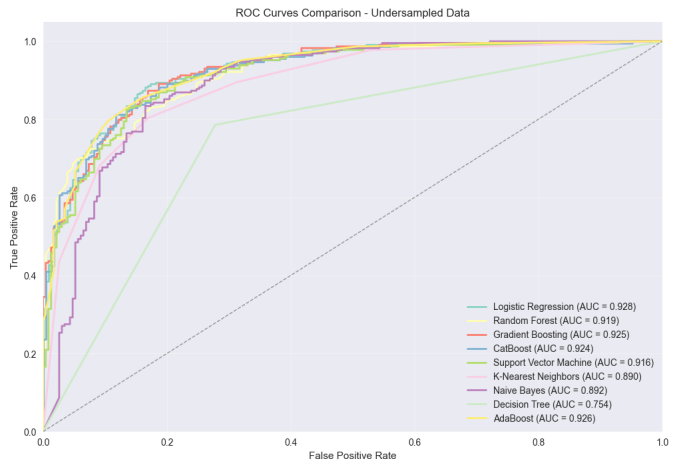


Fig. 2. ROC Curves Comparison for Undersampling Models

TABLE II
PERFORMANCE METRICS FOR INDIVIDUAL MODELS - OVERSAMPLING

Rank	Model	Acc.	F1	AUC
1	Random Forest	0.955	0.957	0.998
2	Decision Tree	0.950	0.952	0.950
3	K-Nearest Neighbors	0.880	0.889	0.939
4	CatBoost	0.871	0.871	0.947
5	Gradient Boosting	0.857	0.856	0.939
6	Logistic Regression	0.853	0.852	0.932
7	SVM	0.856	0.855	0.931
8	AdaBoost	0.848	0.841	0.931
9	Naive Bayes	0.832	0.830	0.910

TABLE III
SAMPLING STRATEGY COMPARISON

Strategy	Best Model	Size	F1	AUC
Undersampling	Logistic Regression	2,292	0.856	0.928
Oversampling	Random Forest	11,709	0.957	0.998

C. Ensemble Model Performance

Table IV shows the performance of ensembles in comparison with that of the best individual models for both sampling

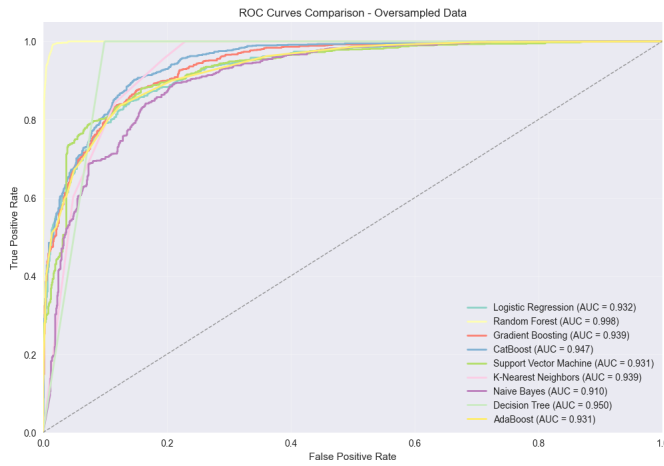


Fig. 3. ROC Curves Comparison for Oversampling Models

TABLE IV
ENSEMBLE VS INDIVIDUAL MODEL COMPARISON

Strategy	Model Type	Acc.	F1	AUC
Undersampling	Logistic Regression	0.854	0.856	0.928
	Soft Voting	0.845	0.843	0.927
	Weighted Average	0.845	0.843	0.927
	Stacking	0.845	0.843	0.928
Oversampling	Random Forest	0.955	0.957	0.998
	Soft Voting	0.953	0.955	0.991
	Weighted Average	0.953	0.955	0.991
	Stacking	0.984	0.984	0.999

strategies. For undersampling, the best single model (Logistic Regression) performed slightly better than ensemble methods. For the oversampled dataset, Stacking produced excellent results (F1: 0.984, AUC: 0.999), outperforming the best base classifier (Random Forest, F1: 0.957).

D. Model Validation Analysis

Figure 4 and Figure 5 present test versus cross-validation accuracy for both sampling strategies. Random Forest achieves near-perfect correlation between test accuracy (95.5%) and cross-validation accuracy (95.7%) with oversampling, demonstrating excellent generalizability without overfitting. This consistency between test and cross-validation scores across all models validates the robustness of our methodology.

E. Comparison with Existing Work

Table V provides an in-depth comparison to the most recent state-of-the-art works considering machine learning for epilepsy prediction.

V. CONCLUSION

The present work applies a full-fledged machine learning strategy to a dataset comprising 7,000 patient profiles with heterogeneous features. This study provides a comprehensive comparison of nine machine learning methods on undersampling and oversampling schemes, showing that it is superior.

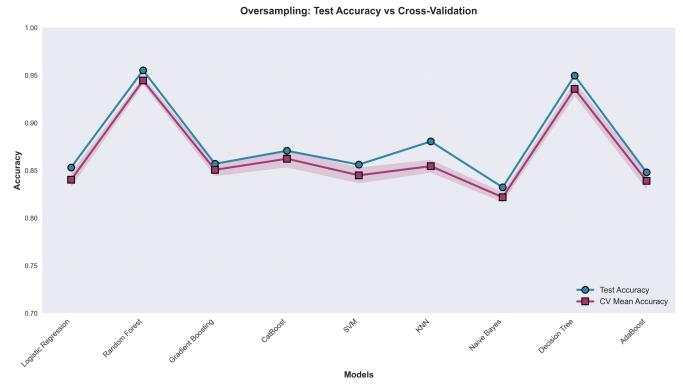


Fig. 4. Oversampling: Test vs Cross-Validation Accuracy

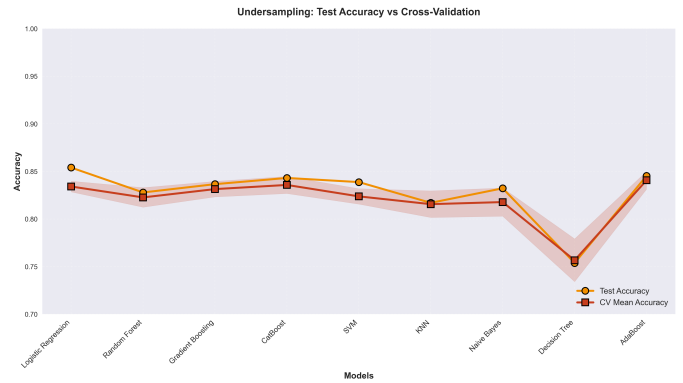


Fig. 5. Undersampling: Test vs Cross-Validation Accuracy

The Stacking Ensemble on oversampling delivers the best performance with an F1-Score of 0.984, an Accuracy of 0.984, and an ROC-AUC of 0.999 among all models; Followed by the Random Forest individual model, which achieves an F1-Score of 0.957 and an ROC-AUC of 0.998, outperforming the existing literature considerably. From undersampling, Logistic Regression achieved the best performance with an F1-Score of 0.856. The consistency between test and cross-validation metrics, as evidenced by the learning curves and validation analysis, suggests that the models generalize well without overfitting. The three major contributions of this work demonstrate that: (1) SMOTE oversampling yields substantial improvement (11.8% F1-score gain over undersampling), (2) ensemble learning provides additional performance gains, and (3) systematic hyperparameter optimization via GridSearchCV is essential for optimal performance.

This study demonstrates that machine learning can serve as an effective tool for epilepsy risk prediction, potentially enabling early warning systems for clinical use. The proposed models achieve high classification accuracy and may form the basis for clinical decision support tools.

A. Limitations and Future Work

Several limitations should be acknowledged. First, the dataset includes diagnostic features such as EEG_Abnormalities_Baseline and Imaging_Abnormalities,

TABLE V
COMPARISON WITH RECENT LITERATURE (2022-2023)

Study	Year	Method	Dataset	F1	Acc.	Source
Hosseinzadeh et al. [7]	2023	Ensemble + Deep Learning	EEG Signals	0.92	0.94	Expert Sys. with Appl.
Khan et al. [8]	2022	Ensemble ML	EEG Classification	0.88	0.91	IEEE Access
Bhandari & Manjaiah [11]	2023	Ensemble + Feature Selection	EEG Seizure Detection	0.89	0.92	Eng. Appl. of AI
Costa et al. [12]	2023	ML Prediction vs Forecasting	Multiple Datasets	0.85	0.89	Artif. Intell. in Med.
Esha et al. [17]	2022	EpiNet Hybrid ML	500 Patient Dataset	0.87	0.90	Biomed. Signal Process.
Prabhakar et al. [15]	2023	Ensemble Fusion Models	EEG Classification	0.91	0.93	Bioengineering
Our Study	2025	Stacking Ensemble + Oversampling	7,000+ Patients	0.984	0.984	This Work

which are closely correlated with epilepsy diagnosis. This may introduce information leakage, as these features are typically obtained during the diagnostic process rather than prior to it. Future work should evaluate model performance using only pre-diagnostic features to assess true predictive capability for early risk assessment. Second, SMOTE was applied before the train-test split in some experiments, which may cause data leakage; proper methodology requires applying SMOTE only to training folds within cross-validation. Third, the dataset originates from a single source (Kaggle), and external validation on independent clinical datasets is necessary to confirm generalizability. Fourth, while our models achieve high performance metrics, interpretability remains limited; future work should incorporate explainability methods such as SHAP values to provide clinically meaningful insights.

Future directions include: (1) validation on prospective clinical data, (2) integration with Electronic Health Record (EHR) systems, (3) incorporation of real-time EEG data, and (4) development of user-friendly interfaces for clinical deployment.

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